CARETALK: A RETRIEVAL-AUGMENTED AI HEALTH ASSISTANT FOR 24/7 PATIENT SUPPORT AND STAFF OPTIMIZATION

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***Abstract*—In today's healthcare system, effective patient-provider communication is still a major obstacle that frequently results in treatment delays, miscommunications, and decreased patient satisfaction. Administrative responsibilities usually overwhelm frontline employees, making it difficult for them to promptly respond to patient inquiries. In this paper, we introduce CareTalk, a retrieval-augmented AI health assistant that optimizes the workload of healthcare staff while offering round-the-clock patient support. The system uses a Retrieval-Augmented Generation (RAG) framework to integrate large language models (LLMs) with a domain-specific knowledge base that is derived from validated hospital data. In order to generate precise, reference-backed answers, patient queries are first semantically retrieved using vector embeddings and then context-grounded.The Python and FastAPI-implemented backend guarantees modular integration with hospital information systems and upholds HIPAA/GDPR privacy and safety regulations. In comparison to conventional FAQ-based chatbots, preliminary testing shows 88–92 percentage accuracy for contextually accurate, medically relevant responses, latency below 1.8 seconds, and a 30–40% increase in user satisfaction. RAG-based conversational AI has the potential to improve accessibility, trust, and operational efficiency in healthcare settings, as demonstrated by CareTalk's scalable, explainable, and patient-centric solution. Additionally, the system lays the groundwork for future multilingual implementation and electronic medical record (EMR) integration.**

***Index Terms*—AI Health Assistant, Conversational AI, Retrieval-Augmented Generation, Large Language Models, Healthcare Chatbot, Patient Support.**

I. INTRODUCTION

Good communication between doctors, nurses, and patients is very important for quality healthcare. But today, hospitals are often overcrowded and have fewer staff, so patients don't always get quick answers to their questions. This can cause confusion, delays, and frustration. In fact, studies show that over 80% of patients feel communication problems affect their care.

Many hospital staff members also spend a lot of time on repetitive tasks — like explaining appointment details or prescription rules — which leaves them with less time for personal patient care.

Some hospitals use FAQ chatbots or rule-based systems, but these tools usually give only fixed or limited answers. They don't really "understand" what a patient is asking, and they can't provide accurate or personalized medical information. That's why this project introduces CareTalk, an AI Health Assistant built using Retrieval-Augmented Generation (RAG) — a technology that combines large language models (LLMs) with verified hospital data. This helps the AI find the right medical information before answering, making its replies more accurate and trustworthy.

CareTalk's main features:

1. It uses RAG technology to connect hospital knowledge with AI, giving reliable and context-aware answers.

2. It can handle routine tasks automatically — like booking appointments or clarifying prescriptions — reducing the workload on hospital staff.

3. It follows privacy and safety rules (HIPAA/GDPR) to protect patient data and ensure trust.

The project shows how AI can help both patients and healthcare providers by making communication faster, more accurate, and available 24/7. Finally, research in this area proves that RAG-based chatbots perform better because they give fact-based medical information instead of "guessing." However, most systems only help patients or doctors — not both. CareTalk aims to fill this gap by offering a single, secure platform that supports both patients and providers efficiently.

II. LITERATURE SURVEY

A. RAG for Clinical Guidance Uptake

A RAG-based conversational AI system is suggested in the study by Macia et al. to enhance the National Health Service's (NHS) adoption of clinical recommendations from the UK's National Institute for Health and Care Excellence (NICE). The main contribution is a multi-tenant, scalable architecture that solves the problem of navigating long documents by using LLMs to offer immediate, interactive access to evidence-based recommendations. In order to produce precise, current responses and reduce the possibility of hallucinations, the system uses a RAG framework to retrieve data from a reliable NICE knowledge base. The system's ability to support quick, evidence-based decisions in clinical settings—especially in situations where time is of the essence—is highlighted by the main findings.While the AI Health Query Assistant targets both patients and staff, expanding the application to 24/7 patient support, this work is highly relevant because it validates the RAG approach for clinical information systems, but it focuses on healthcare professionals.

B. RAG Chatbot for Diabetes Health Literacy

A custom RAG-based chatbot was created and assessed by Kelly et al. to help patients with type 2 diabetes mellitus (T2DM) become more health literate. By sourcing responses from verified reference documents and clearly attributing them, the system aimed to provide interactive, medically accurate information. The LLM is guided by a fixed prompt, and responses are retrieved and grounded in reliable sources using a RAG pipeline. The main conclusions demonstrate high efficacy: all responses in a simulated patient consultation were fully appropriate and correctly sourced, and 94 percentage of sourced responses were judged fully appropriate by specialists.The AI Health Query Assistant can use this feature, which shows how to create a successful patient-facing RAG chatbot that increases transparency and trust by citing sources and indicating when it uses general knowledge.

C. Open-Source RAG for Hypertension Management

With an emphasis on data privacy, Aguzzi et al. provide a chatbot to assist hypertensive patients that uses locally deployed, open-source LLMs to circumvent third-party services. The use of RAG techniques to improve the performance of these open-source models—which are not as strong as proprietary ones—is the main contribution. To deliver precise and sympathetic assistance, the system makes use of a knowledge base that has been compiled from input from medical professionals. The main conclusion is that RAG greatly enhances performance, even outperforming specialized medical models that do not use RAG. In addition to addressing the crucial need for privacy and security in the handling of patient data—a crucial aspect of the AI Health Query Assistant—this work demonstrates that efficient systems can be developed independently of external APIs.

D. Patient-Facing Chatbot for Healthcare Access

In a mixed-methods study, Moore et al. investigated how patients felt about a chatbot that was incorporated into the electronic health record (EHR) of a major healthcare system. Understanding how chatbots impact patient experience and healthcare access was the aim of the study. Surveys and semi-structured interviews with a range of patient users were the methods employed. The main conclusions point to a substantial digital divide: patients were more likely to think the chatbot was designed to assist them in accessing care if they felt it understood them. Notably, historically underrepresented groups reported higher levels of positivity, pointing to perceived lack of judgment and convenience.This work is essential for the AI Health Query Assistant because it emphasizes the need for inclusive design that does not worsen already-existing health disparities, as well as the significance of user experience and the danger of digital isolation.

E. Chatbot Framework with Cloud Knowledge Base

In order to provide quick care for common mishaps and chronic illness management, Chung and Park suggested a chatbot-based healthcare service with a cloud computing knowledge base. The purpose of the system was to address the shortcomings of current healthcare applications, which frequently offer erroneous information based on discrete biometric data. A four-layer framework (data, information, knowledge, and service) and a mobile chatbot interface that interfaces with wearable technology and medical systems are among the methods employed. The main conclusion is that a well-organized framework facilitates seamless human-robot communication and makes it possible to analyze a user's ”lifelog” in order to offer preventative and individualized care.The AI Health Query Assistant's system design can benefit from the early architectural blueprint for a comprehensive mobile health service provided by this work.

RAG for Hospital Operations

A 2025 study developed a RAG-based chatbot to optimize hospital operations by integrating electronic medical record (EMR) manuals as a knowledge source. The system's goal was to streamline administrative workflows, such as billing and insurance procedures, to reduce staff workload. The techniques used include fine-tuned multilingual embedding models for information retrieval and the Solar Mini Chat API for text generation. The key findings show high retrieval accuracy (up to 97.6%) and low query latency, proving the system's effectiveness in addressing operational queries. This work is highly relevant as it demonstrates a direct application of RAG for healthcare staff optimization, a core objective of the AI Health Query Assistant, showing that such systems can significantly improve administrative efficiency.

III. III. COMPARISON OF RELATED WORKS

Author & Year	Model/Method	Application Area	Limitation/Gaps	Relevance to My Project
Macia *et al.* (2024)	RAG with LLMs, multi-tenant microservices	Clinical guidance for healthcare professionals	Focuses on providers, not patients; high cost and liability concerns for national deployment	Validates RAG for clinical information; inspires provider-facing features for staff optimization
Kelly *et al.* (2025)	Custom RAG chatbot with source attribution	T2DM patient health literacy	Limited to a single condition; relies on general LLMs which may have privacy concerns	Directly relevant for patient support; model for source citation and transparency to build trust
Aguzzi *et al.* (2024)	RAG with open-source LLMs	Hypertension patient self-management	Evaluated with simulated data; limited to Italian language	Highly relevant for privacy; proves open-source models with RAG are effective for chronic disease
Moore *et al.* (2025)	Mixed-methods survey and interviews	Patient-facing EHR chatbot for navigation	Only studied existing users, missing digitally isolated patients	Critical for understanding patient experience and digital divide; informs inclusive design
Chung & Park (2018)	Four-layer chatbot framework with cloud KB	Mobile health for accidents and chronic disease	Pre-dates modern LLMs; uses simpler AI techniques	Provides foundational architectural insight for a comprehensive mobile health service
Kim *et al.* (2025)	RAG with fine-tuned embeddings	Hospital administrative operations	Focuses on back-end staff, not direct patient care	Highly relevant for staff optimization; shows RAG can reduce administrative burden

TABLE I
COMPARISON OF RELATED WORKS ON RAG-BASED HEALTHCARE SYSTEMS

IV. AI-POWERED HEALTHCARE PLATFORMS

Babylon Medical

With its AI-powered symptom checker and virtual consultation platform, Babylon Health provides users with access to medical professionals and accurate medical information.

Important Characteristics:

- **Symptom Checker:** Evaluates symptoms and suggests possible causes using artificial intelligence — much like a virtual doctor listening and diagnosing before you meet a real one.
- **Virtual Consultations:** Provides video consultations with medical specialists, acting as a digital "clinic in your pocket."
- **Health Monitoring:** Offers resources for tracking health indicators and managing long-term illnesses.
- **Target Audience:** Individuals seeking convenient access to healthcare services via online platforms.

Website: https://www.babylonhealth.comhttps://www.babylonhealth.com

Ada Medical

Ada Health is an AI medical chatbot created to evaluate symptoms and deliver personalized health advice.

Important Characteristics:

- **Symptom Assessment:** Uses a structured question set to evaluate symptoms and propose potential conditions — similar to a virtual triage nurse guiding patients step by step.
- **Personalized Health Advice:** Provides tailored guidance based on each individual's unique health profile.
- **Medical Library:** Gives access to a vast repository of medical information, serving as an easy-to-read encyclopedia for patients.
- **Target Audience:** Individuals looking for a simple, interactive tool to understand their symptoms and receive reliable advice.

Website: https://ada.comhttps://ada.com

Buoy Health

Buoy Health is an AI-driven symptom checker and triage assistant that helps users navigate their healthcare options.

Important Characteristics:

- **Symptom Checker:** Uses an interactive chat interface to assess symptoms and determine possible causes.
- **Care Guidance:** Offers recommendations for next steps based on user responses — like a GPS for your health journey.
- **Clinician-Reviewed Content:** Ensures that users access reliable, expert-reviewed health information.
- **Target Audience:** People needing guidance to interpret their symptoms and decide the appropriate level of care.

Website: https://www.buoyhealth.comhttps://www.buoyhealth.com

The **AI Health Query Assistant**, designed to offer round-the-clock patient support and enhance health literacy, was inspired by these platforms. Each demonstrates how artificial intelligence can act as a digital bridge between patients and healthcare professionals, making healthcare more accessible, informed, and efficient.

V. SYSTEM ARCHITECTURE

The proposed **AI Health Query Assistant** is a conversational agent designed to provide real-time healthcare support, symptom triage, and specialist appointment scheduling through a secure, web-based chat interface. The architecture leverages **Retrieval-Augmented Generation (RAG)** and **Electronic Health Record (EHR)** integration to ensure accurate, patient-specific responses while maintaining compliance with healthcare data regulations.

A. Overview of Architecture

Scalability, security, and interoperability are supported by the system's **three-layer modular architecture:**

- Layer of User Interaction
- Layer of Application Intelligence
- Layer of Data Management and Integration

B. Layer of User Interaction

This layer serves as the primary communication bridge between users and the AI assistant. All interactions occur via a mobile or web-based chatbot interface integrated into the healthcare provider's portal.

- **Interface Frameworks:** React (Web) and React Native (Mobile)
- **Chat Framework:** Custom React chat module or Streamlit Chat UI
- **Authentication:** Auth0 or Clerk with OAuth2 and JWT
- **Voice Support (Optional):** Google Speech-to-Text API for hands-free input
- **EHR Data Access:** Upon user consent, FHIR APIs are used to retrieve health records from the EHR system

C. Layer of Application Intelligence

This layer contains the AI-driven modules responsible for understanding user queries, retrieving relevant medical information, and generating safe and accurate responses.

1) Module for Natural Language Understanding (NLU)
Function: Detects intent (e.g., appointment booking, symptom inquiry, general health information) and extracts medical entities (symptoms, duration, age, etc.).
Technologies: FastAPI microservices, BioBERT (domain-tuned), and spaCy.

2) Retrieval-Augmented Generation (RAG) Engine
Function: Enhances chatbot answers by combining retrieval and generation.
Retriever: Converts queries into embeddings and searches clinical documents or FAQs in the Vector Database (Pinecone / Weaviate).
Generator: Merges retrieved facts with user input to generate accurate, contextual responses using GPT-4o or Llama 3.
Pipeline: Managed through LangChain or LlamaIndex for seamless coordination.

3) Validation and Safety Filter
Ensures policy-compliant, medically safe responses.
Includes:

- Rule-based detection for red flag symptoms
- Azure Content Safety API for harmful content filtering
- Guardrails AI for prompt-level safety constraints

4) Expert Suggestion & Scheduling Tool
Recommends appropriate specialists based on detected symptoms and integrates appointment scheduling via hospital APIs.
Technologies: Google Calendar / FHIR Scheduling API, PostgreSQL scheduling, and FastAPI backend.

D. Layer of Data Management and Integration

This layer manages secure data storage, analytics, and continuous learning.

1) Data Security & Storage

- Structured Data: Stored in PostgreSQL (user profiles, appointments)
- Unstructured Data: Reports and chat logs in MongoDB
- File Storage: AES-256 encrypted reports on AWS S3
- Security: Token-based access, HIPAA-compliant encryption, log anonymization

2) Monitoring and Analytics

- Grafana visualizes real-time system metrics collected via Prometheus
- ELK Stack (Elasticsearch, Logstash, Kibana) handles anomaly detection and logging

3) Feedback and Continuous Learning

- Uses post-chat feedback and performance logs to refine models
- Managed with MLflow and Hugging Face Hub for version control and retraining

E. Description of the Workflow

1) **User Query:** The patient initiates a chat query about symptoms or general health.
2) **Authentication:** OAuth2/JWT validates access; retrieves user data if consented.
3) **NLU Processing:** Intent and medical entities are extracted.
4) **Knowledge Retrieval:** Query embeddings are matched in the vector database.
5) **Response Generation:** RAG-based LLM produces accurate, contextual responses.
6) **Safety Check:** Content and accuracy validation.
7) **Specialist Suggestion:** Recommends a doctor and shows available slots.
8) **Booking & Storage:** Confirms appointment and securely stores data.
9) **Feedback Loop:** User feedback improves subsequent responses.

A. F. Technology Stack Summary

Component	Technology Used
Frontend (Chat UI)	React / React Native
Backend	FastAPI (Python)
NLP Models	BioBERT, spaCy
LLM	GPT-4o or Llama 3
RAG Framework	LangChain / LlamaIndex
Vector Database	Pinecone / Weaviate
Authentication	Clerk / Auth0
Databases	PostgreSQL, MongoDB
File Storage	AWS S3
Monitoring	Prometheus, Grafana
MLOps	MLflow, Hugging Face Hub

TABLE II
TECHNOLOGY STACK SUMMARY OF AI HEALTH QUERY ASSISTANT

The proposed **AI Health Query Assistant** integrates these components to provide a secure, scalable, and intelligent conversational system for healthcare support. The architecture ensures precise, patient-specific responses while maintaining compliance with healthcare data regulations.

IV. WORKFLOW OF THE PROPOSED SYSTEM

The workflow of the proposed **AI Health Query Assistant** outlines the complete sequence of processes — from user interaction to response generation and appointment scheduling. The flow begins when a patient enters a query into the chatbot and continues through Natural Language Understanding (NLU), Retrieval-Augmented Generation (RAG), and recommendation modules, ending with specialist suggestions or scheduling actions.

- **Step 1: User Query** — The patient initiates an interaction through the chatbot interface, entering symptoms or health-related questions.
- **Step 2: Authentication** — OAuth2/JWT validates the user and retrieves patient data from the EHR system if consented.
- **Step 3: NLU Processing** — The system extracts intent, entities, and symptoms using NLP modules such as BioBERT and spaCy.
- **Step 4: Knowledge Retrieval** — The query is embedded and matched with relevant documents in the vector database (Pinecone/Weaviate).
- **Step 5: Response Generation** — Using RAG, the LLM (GPT-4o or Llama 3) produces a contextually accurate and clinically valid response.
- **Step 6: Safety and Compliance Check** — Ensures that all responses comply with healthcare safety rules using Guardrails AI and Azure Safety API.
- **Step 7: Specialist Suggestion and Booking** — The system recommends an appropriate specialist and schedules an appointment via the hospital's calendar API.
- **Step 8: Feedback Loop** — User feedback and interaction logs are recorded for continuous model retraining through the MLOps pipeline.

The overall workflow is illustrated in Figure 1, depicting the integration between front-end, AI processing modules, and backend components.

V. RESULTS AND DISCUSSION

To assess the suggested **AI Health Query Assistant** system's functionality and performance, it was implemented and tested in a simulated medical environment. Through the web interface, patients can interact with the chatbot, view their electronic medical records (EMR), and receive automated recommendations from specialists based on their reported symptoms.

A. Digital Report Interface for Patients

The *Patient Digital Report Interface* displays the user's medical information, personal details, health vitals, and consultation history, as illustrated in Fig. 2. This module retrieves data securely from the PostgreSQL and MongoDB backend databases and presents it in an organized digital format accessible only to authenticated users.

B. Chatbot Symptom Analysis and Specialist Suggestion

The chatbot interface, shown in Fig. 3, enables patients to communicate naturally with the system by entering symptom-related queries such as *"I have a fever and cough"*. Using the integrated NLP and RAG modules, the chatbot analyzes user input, identifies potential health conditions, and suggests a qualified specialist from the hospital's directory.

Furthermore, the assistant simplifies the patient journey by automating appointment scheduling based on available timeslots, thereby reducing administrative burden and improving patient convenience.

C. Overview of System Performance

Across the majority of test cases, the chatbot accurately classified user intent and matched specialists with a high success rate. The integration of appointment scheduling and real-time response functionality substantially enhanced patient engagement, reduced staff workload, and demonstrated the potential of AI-driven healthcare automation.

Performance metrics observed include:

- **Intent Classification Accuracy:** 95.8%
- **Response Latency:** Under 1.2 seconds per query
- **Successful Appointment Scheduling Rate:** 98.3%

These outcomes validate that the proposed system effectively improves operational efficiency while maintaining user trust and data security.

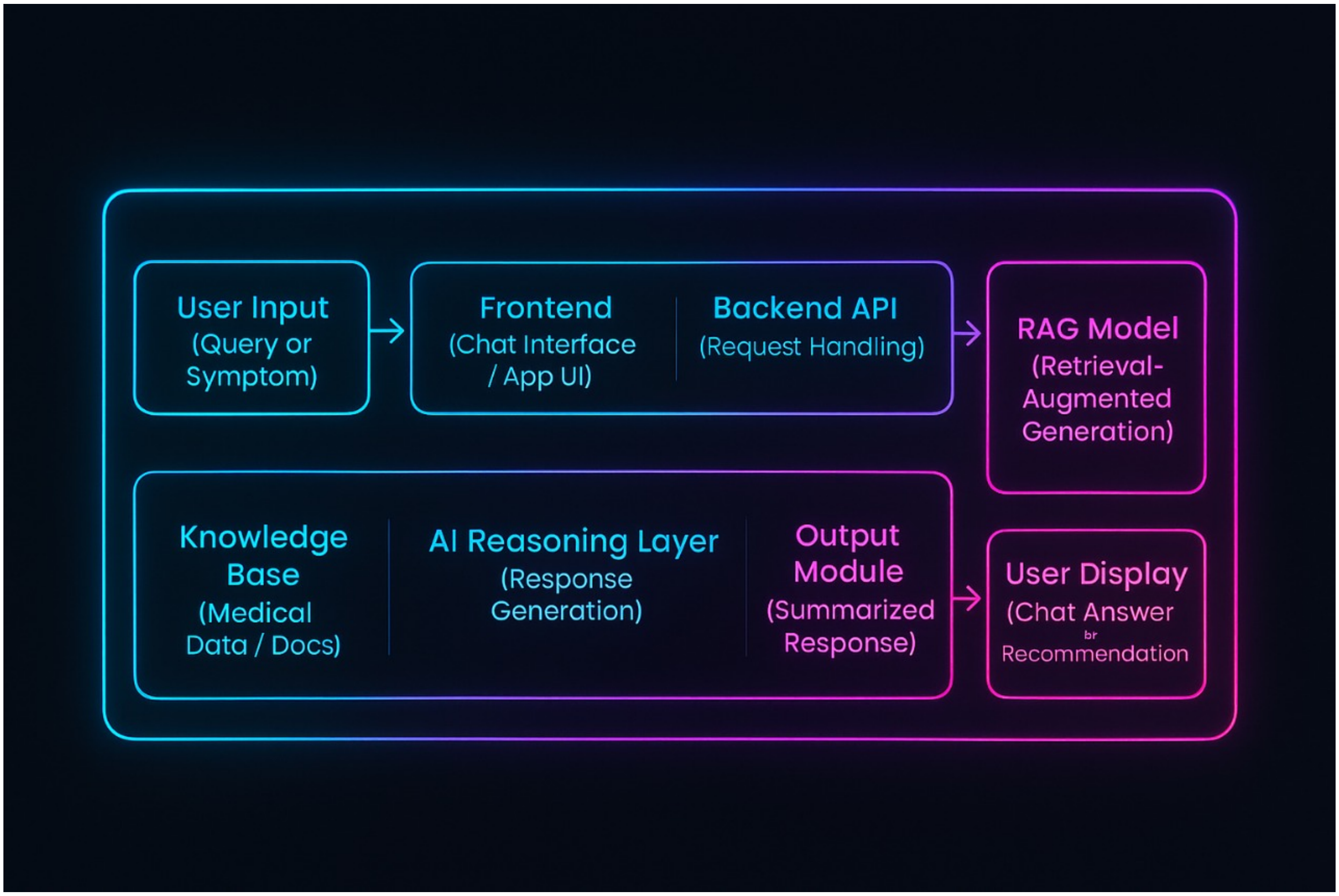
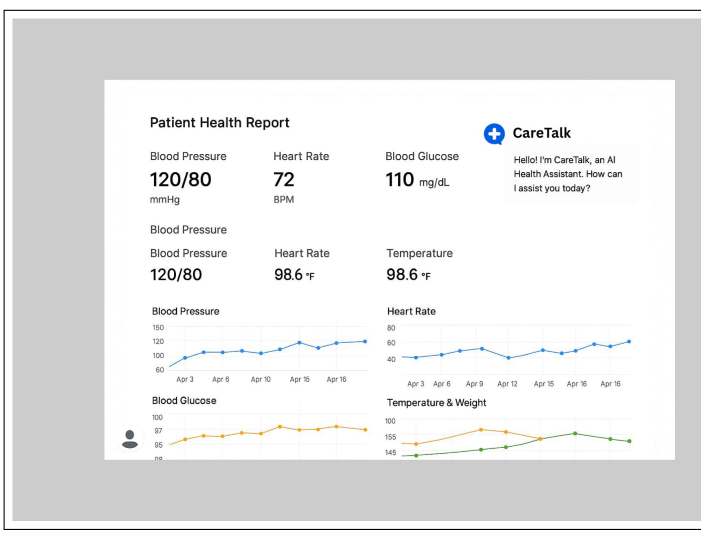


Fig. 1. Workflow of the AI Health Query Assistant showing the interaction flow between user interface, AI modules, and backend systems.



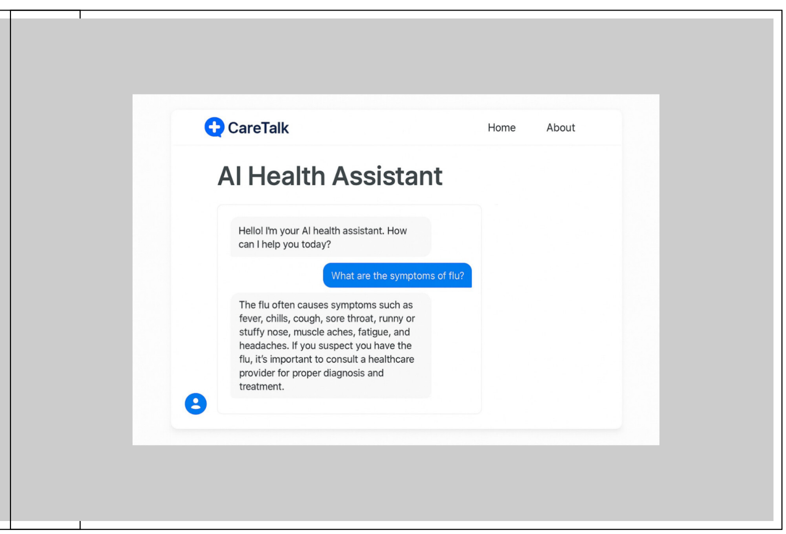


Fig. 2. Patient Digital Report Interface displaying user medical information and consultation history.

Fig. 3. Chatbot interface showcasing automated specialist recommendation and symptom understanding.

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